

E0196

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Statistics for Business Administration

Exam: MA9712 / Retake-MA9712-22

Date: Tuesday 4th October, 2022

Examiner: Prof. Mathias Drton

Time: 08:15 – 09:45

Working instructions

- This exam consists of **16 pages** with a total of **10 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 44 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **scientific pocket calculator**
 - the **cheat sheet**, which is contained in this exam on page 14
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____

Problem 1 (5 credits)

Give a short answer to each of the following sub-problems.

- 0 ☐ 1 ☐ a) A political scientist is interested in the effect of government type on economic development. She wants to use a sample of 30 countries evenly represented among North America, South America, Europe, Asia, and Africa to conduct her analysis. What type of study and sampling strategy should she use to ensure that countries are selected from each region of the world?

Hint: Your answer should include at least two of the terms: observational, experimental, simple random sample, cluster sample, stratified sample

She will only be able to conduct an **observational study**. The five continents represent five different strata and from each stratum a random sample should be drawn. Hence, it should be a **stratified sample**

- 0 ☐ 1 ☐ b) Consider the following card game with a well-shuffled deck of 52 cards. If you draw a red card, you win nothing. If you get a spade, you win 5\$. For any club, you win 10\$ plus an extra 30\$ for the ace of clubs. Find the expected winnings for a single game.

The random variable X describes the random outcome of the game, which has sample space $S = \{0, 5, 10, 40\}$. Then we need to compute

$$E(X) = \sum_{x \in \{0, 5, 10, 40\}} x \cdot P(X = x) = 0 \cdot \frac{26}{52} + 5 \cdot \frac{13}{52} + 10 \cdot \frac{12}{52} + 40 \cdot \frac{1}{52} \approx 4.33.$$

- 0 ☐ 1 ☐ c) Suppose in a population 20% of adults do not have a savings account. What is the expected shape of the sampling distribution of proportions of adults without a savings account in random samples of 60 adults from this population? Justify your answer.

The shape of the distribution is supposed to be symmetric, which is due to the CLT. The conditions of the CLT are fulfilled, since we have independent sample observations and a sufficiently large (≥ 30) sample size.

- 0 ☐ 1 ☐ d) Consider the following statement: "At the 95% confidence level, we estimate that the true population proportion of Molwanen teenagers that smoke cigarettes is at least 0.15 but no more than 0.24. If we were to construct a confidence interval at the 99% confidence level, the resulting interval would be narrower." Decide if there are any errors in the statement. If so, **correct** the statement.

The interval would be **wider** instead of narrower.

- 0 ☐ 1 ☐ e) Researchers wish to find out whether a new treatment for depression is effective. Depression was assessed by the Beck's depression inventory, which results in numerical scores that range from 0 to 63. Which of the methods introduced in the lecture is the most appropriate hypothesis test to determine if depression scores differ pre- and post-treatment?

The data will consist of paired observations. Since we need to analyse numerical data, it has to be a **paired t-test**.

Problem 2 (2 credits)

Suppose the true proportion of students at a college who study abroad is 0.3. Imagine you select a random sample of 40 students from the college and record if they have studied abroad.

- a) Let X_i be the random event *student i studies abroad*. What is the distribution of the random variable $\sum_{i=1}^{40} X_i$?

The event <i>student i studies abroad</i> is a random event, which can be described by a Bernoulli(p) distributed random variable X_i . Hence, the sum $X = \sum_{i=1}^{40} X_i$ over all 40 students is a Binomial(40,0.3) distributed random variable.																																							
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- b) What is the probability that at most a quarter of the students has studied abroad?

Hint:

```
pgeom(5:15, prob = 0.3)
## [1] 0.882 0.918 0.942 0.960 0.972 0.980 0.986 0.990 0.993 0.995
## [11] 0.997

pbinom(5:15, size = 40, prob = 0.3)
## [1] 0.009 0.024 0.055 0.111 0.196 0.309 0.441 0.577 0.703 0.807
## [11] 0.885

dbinom(5:15, size = 40, prob = 0.3)
## [1] 0.006 0.015 0.032 0.056 0.085 0.113 0.132 0.137 0.126 0.104
## [11] 0.077
```

The probability that at most a quarter of the students (=10) studies abroad, is the probability that $P(X \leq 10)$. This means, we have to use the pbinom() output, to get																																							
$P(X \leq 10) = 0.309$.																																							

Problem 3 (6 credits)

In a survey of Family Growth researchers gather information on family life, marriage and divorce, pregnancy, infertility, use of contraception, and men's and women's health (contained in the object afc). One of the variables collected in this survey is the age of women at the birth of the first child. 5,534 randomly sampled US women between 2006 and 2010 completed the survey. The women sampled here had at least one child.

```
afc %>%
  summarize(mean(age), median(age), sd(age))
## # A tibble: 1 x 3
##   `mean(age)` `median(age)` `sd(age)`
##   <dbl>      <dbl>      <dbl>
## 1      25.4        25      4.72
```

The researchers would like to show that the mean age of first-time mothers is less than **25.5 years**.

- a) Specify null and alternative hypothesis based on the above problem statement.

We want to show that the mean age is smaller than 25.5. Therefore, the null and alternative hypothesis are given by																																							
$H_0 : \mu = 25.5 \quad H_A : \mu < 25.5$.																																							

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b) Make a guess about statistical significance in the testing problem based on the given results in the problem statement.

We observed that the sample mean 25.4401879 is just slightly below the null hypothesis 25.5. We also saw from the boxplot that the variation is not huge, but also not really small, since $\text{sd}(\text{afc})$ is equal to 4.721365. Hence, we might not be able to reject H_0 . But we shouldn't forget that the sample size is large, which reduces the standard deviation of the sample mean and maybe allows us to reject H_0 .

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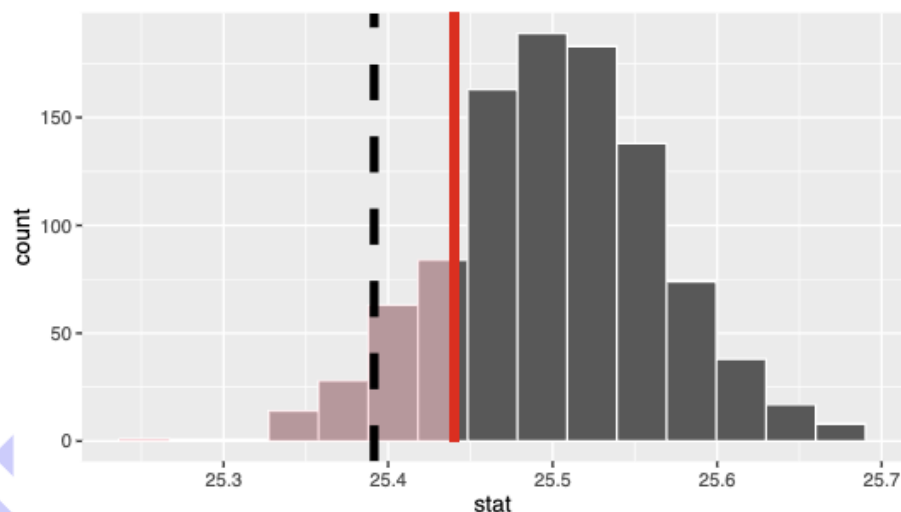
c) Use the following results

```
library(infer)null_dist <- afc %>%
  specify(response = age) %>%
  hypothesize(null = "point", mu = 25.5) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "mean")
```

```
qstat <- quantile(null_dist$stat, probs = 0.05)
```

```
null_dist %>%
  visualize() +
  shade_p_value(obs_stat = mean(afc$age), direction = "less") +
  geom_vline(xintercept = qstat, size = 2, linetype = 2)
```

Simulation-Based Null Distribution



to test the considered hypotheses at a significance level of 0.05. Decide if you can reject the null hypothesis. Justify your answer.

The plot contains the p-value as a shaded area. It already looks larger than the assumed significance level of 5%. But the plot contains additionally the 0.05 quantile of the simulated test statistic values. Since the p-value should be less than the significance level, the solid line should be left to the dotted line, which is not the case. Hence, the data does not show enough evidence to reject the null hypothesis. This means, that it is not unlikely (enough) to observe an empirical mean of 25.4401879 or less than that, if the true mean value is assumed to be 25.5 years.

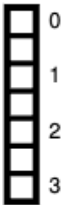
Problem 4 (3 credits)

A random sample of 600 24-35 year-old unemployed Americans yielded an average unemployment of 13 weeks. In order to construct a bootstrap confidence interval based on this sample a statistician took 100 bootstrap samples and recorded their means. The standard deviation of these means was found to be 0.5.

Compute the 95% bootstrap confidence interval using the standard error method.

Hint:

```
qnorm(c(0.025, 0.05, 0.95, 0.975))
## [1] -1.96 -1.64 1.64 1.96
```



The interval will be computed using the formula $\bar{x}^* \pm z_{0.975} \cdot SE_{\bar{x}^*}$, where $\bar{x}^*$ is the empirical mean of the bootstrap means, $z_{0.975}$ is the 0.975 quantile of the standard normal distribution and $SE_{\bar{x}^*}$ is the standard deviation of the bootstrap means, hence equal to 0.5.

By construction of the bootstrap method, the expected value of the bootstrap means $\bar{x}^i$, $i = 1, \dots, 100$, will be the empirical mean $\bar{x}$ of the sample values. The problem says, that the standard deviation of the bootstrap means is 0.5 and 0.95 quantile of the $N(0,1)$ distribution is

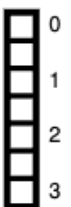
```
qnorm(0.975)
## [1] 1.96
```

Hence, the interval is equal to $13 \pm 1.96 \cdot 0.5 = (12.02, 13.98)$.

Problem 5 (3 credits)

Consider a hypothesis test about two proportions, such as $H_0 : p_1 = p_2$ vs $H_A : p_1 \neq p_2$. Describe how a decrease of the sample sizes n_1 and n_2 affects the absolute value of the test statistic and the corresponding p-value, while assuming, that all parameter estimates stay the same. Use the appropriate formulas for your argument.

Describe how a decrease of the sample sizes n_1 and n_2 affects the absolute value of the test statistic and the corresponding p-value, while assuming, that all parameter estimates stay the same. Use the appropriate formulas for your argument.



Decreasing the sample sizes n_1 and n_2 will increase the standard error, due to fact that

$$SE_0 = \sqrt{\hat{p}_{pooled}(1 - \hat{p}_{pooled}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)},$$

which will lead to a **decrease** of the absolute value of the test statistic, since

$$|z| = \left| \frac{(\hat{p}_1 - \hat{p}_2) - 0}{SE_0} \right|$$

and therefore to an **increase** of the p-value, since

$$P_{H_0}(|Z| > z_1) < P_{H_0}(|Z| > z_2)$$

for $z_1 > z_2$.

Problem 6 (8 credits)

The American National Election Studies (ANES) collects data on voter attitudes and intentions as well as demographic information. In this question we will focus on the two variables from the 2012 ANES dataset:

- region (levels: Northeast, North Central, South, and West),
- whether the respondent feels things in this country are generally going in the right direction or things have pretty seriously gotten off on the wrong track.

To keep calculations simple we will work with a random sample of 500 respondents from the ANES dataset. The distribution of responses is as follows:

```
addmargins(table(anes))
##               region
## direction    north_central northeast south west Sum
## right_direction         44         29         62         36 171
## wrong_track             77         54        131         67 329
## Sum                   121         83        193        103 500
```

According to the 2010 Census, 18% of US residents live in the Northeast, 22% live in the North Central region, 37% live in the South, and 23% live in the West.

Given the data, the following statistical analysis was done in R.

```
chisq.test(table(anes$region), p = c(0.22, 0.18, 0.37, 0.23))
##
## Chi-squared test for given probabilities
##
## data:  table(anes$region)
## X-squared = 3, df = 3, p-value = 0.4
```

a) State the null and alternative hypothesis corresponding to the test done in the statistical analysis of the data.

We want to compare the sample distribution with the census distribution. This means, that we want to conduct a χ^2 GOF test. The corresponding hypotheses are

H_0 : The sample distribution of regions follows the census distribution.

H_A : The sample distribution of regions does not follow the census distribution.

b) Check the conditions of the test done in the statistical analysis.

We have to check the following conditions:

1. Independence: The sample is random, hence we can assume that one individual in the sample is independent of another.
2. Sample size: The expected counts can be calculated as follows. We take the data from the 2010 Census and multiply it by the sample size

```
prob <- c(0.22, 0.18, 0.37, 0.23)
500 * prob
## [1] 110  90 185 115
```

All expected counts are greater than 5, so the condition is fulfilled

c) Assume a significance level of 0.05 for the test. State the conclusion of the test in the context of the data. Justify your answer.

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The test statistic is rather small, which makes the p-value rather large and in particular larger than $\alpha = 0.05$. We therefore fail to reject H_0 . The data do not provide convincing evidence that the sample distribution of regions does not follow the census distribution. Hence, the sample is likely representative since the distribution of sampled individual matches the regional population distribution.

d) Now assume we want to test the following hypotheses

- H_0 : Region and opinion on direction are independent.
 H_A : Region and opinion on direction are dependent.

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Complete the R code below so that this command can be used to test the above hypotheses.

```
chisq.test( )
```

`chisq.test(table(anes))` or `chisq.test(table(anes$region, anes$direction))` or
`chisq.test(anes$region, anes$direction)`

e) Under the null hypothesis of independence between the variables region and direction, the expected counts are equal to

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```
##           region
## direction  north_central northeast south west
## right_direction      41.4      28.4    66 35.2
## wrong_track         79.6      54.6   127 67.8
```

Given these values, do you think the null hypothesis can be rejected or not? Justify your answer.

The expected counts are pretty close to the observed counts. Since the χ^2 statistic sums up the weighted squared differences between the observed and expected counts, the value of the χ^2 statistic will be small. But since one rejects the null hypothesis for large values of the test statistic, we think that this will not be the case for the given data.

Problem 7 (4 credits)

For a t-test in R, we obtained the following output.

```
t = 0.2024, df = 14, p-value = 0.8425
alternative hypothesis: true mean is not equal to 20
95 percent confidence interval:
18.91483 21.31132
sample estimates:
mean of x
20.11307
```

Decide if the following statements are true or false. Justify your **answer**.

Remark: Only correct justifications are awarded with points. Not the decision (true/false).

a) This is a one-sided test.

FALSE: The output says that the alternative hypothesis is equal to $H_A : \mu \neq 20$, which is a two-sided alternative.

b) At the significance level $\alpha = 0.05$ the null hypothesis can't be rejected **because** the sample mean is contained in the 95% confidence interval.

FALSE: The null hypothesis can't be rejected, but the argument is wrong. The sample mean is by construction contained in the interval. The null hypothesis can't be rejected because the interval covers the null value 20.

c) There is a 0.157 probability that the null hypothesis is false.

FALSE: Because *Based on a hypothesis test, we can never make a statement about the probability of the null hypothesis being false (or true). We can just quantify how strong the evidence is for favouring the alternative given the observed data. or 1-0.8425 is the probability of observing data at least as favorable to the null hypothesis as the current data set.*

d) This analysis had a sample size of $n = 14$.

FALSE: The output says that a t-distribution with 14 degrees of freedom has been used to compute the p-value. Since $df=n-1$, the number of observations is equal to 15.

Problem 8 (3 credits)

Rhesus monkeys kept in captivity are given a range of items in their living areas for enrichment, and the managers of these captive groups conduct a variety of experiments to see what setups are most beneficial to the monkeys.

In one of the following three scenarios it would be appropriate to conduct analysis of variance (ANOVA). Decide in which scenario this is the case. Justify your choice by **arguing why ANOVA shouldn't be applied in the other two scenarios**. Name for each of the other two scenarios one of the three conditions (1. independence, 2. normality, 3. homogeneous variability) for ANOVA that is violated in this scenario and say why this is the case.

- i) Researchers divide monkeys into four groups, each group with a different enrichment setup. After one week, blood samples are collected to measure stress hormone levels, and the researchers are interested in seeing if the mean stress levels differ amongst the groups.
- ii) Researchers collect baseline levels of stress hormones from one group of monkeys with no enrichment. After this baseline measurement, researchers add a climbing toy to the living area. After one week, they collect another hormone sample from the same monkeys and want to compare the baseline levels to the post-enrichment levels.
- iii) Researchers collect behavioural data on monkeys and categorize them as either "high stress" or "low stress." They also have information on what their enrichment classification is in their group (food-focused, toy-focused, or structure-focused). They are interested in seeing if there is an association between the monkeys' stress category and enrichment classification.

The ANOVA method can be used for numerical data to check whether the means across many (more than two) independent groups are equal.

- ii) Independence does not hold, since we have **paired** data, which introduces dependence.
- iii) The response variable stress is not nearly normal, since it is **categorical** with levels high and low.

Hence, ANOVA is just appropriate for scenario i).

Problem 9 (5 credits)

Many university courses conclude by giving students the opportunity to evaluate the course and the instructor anonymously. A study examining professor evaluations collected data on student evaluation scores of professors (1-low to 5-high), their rank, department, etc.

In class we evaluated the relationship between professor ratings and the professor's rank (non tenure track, tenure track, tenured). The following are some summary statistics and the ANOVA table associated with this analysis:

```
beauty %>%
  group_by(tenure) %>%
  summarize(q1 = quantile(profevaluation, 0.25), q2 = median(profevaluation),
            mean = mean(profevaluation), q3 = quantile(profevaluation, 0.75),
            sd = sd(profevaluation), n = n())

## # A tibble: 3 x 7
##   tenure      q1      q2 mean      q3      sd      n
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <int>
## 1 non tenure track  3.9  4.4  4.28  4.7  0.498  102
## 2 tenure track    3.7  4.35 4.15  4.6  0.561  108
## 3 tenured        3.8  4.2  4.14  4.6  0.550  253

summary(aov(profevaluation ~ tenure, data = beauty))
##              Df Sum Sq Mean Sq F value Pr(>F)
## tenure         2      1.6    0.795    2.71  0.068 .
## Residuals    460    135.1    0.294
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

a) Compute the sum of squares total SST for the given ANOVA table.

The sum of squares total is equal to

$$SST = SSG + SSE = 1.6 + 135.1$$

b) Suppose the mean score for the non tenure track professors was actually **4.14**. How, if at all, would this change the F score, assuming all other statistics stay the same. Justify your answer

The $F = \frac{SSG/df_G}{SSE/df_E}$ score would decrease, since the sum of square groups $\sum_{i=1}^3 n_i(\bar{x}_i - \bar{x})^2$ would decrease, with $\bar{x}_1$ shifting closer to the other two mean values.

c) Suppose next we want to evaluate whether average professor ratings vary by professors' home departments (levels: 'natural science', 'social science', 'humanities', 'other'). Will the sum of squares total be different in the new ANOVA table? Justify your answer.

Remark: The influence of 'tenure' is not considered anymore in this new analysis.

The sum of squares total measures the total variability of the response values, which is of course unaffected by the choice of the explanatory variable. Hence, SST will not change.

d) Determine the groups and error degrees of freedom for this new ANOVA analysis.

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The variable department has four levels, i.e. the group degrees of freedom are equal to $df_{group} = 4 - 1 = 3$.
Since we have 463 observations, the error degrees of freedom are equal to $df_{error} = 463 - 4 = 459$.

Sample Solution

Problem 10 (5 credits)

The RMS Titanic sank in the early morning hours of 15 April 1912 in the North Atlantic Ocean. The dataset `titanic` contains information about passengers aboard of the RMS Titanic

```
titanic
## # A tibble: 714 x 5
##   Survived Pclass Sex      Age SibSp
##   <int>   <int> <chr>  <dbl> <int>
## 1       0     3 male    22     1
## 2       1     1 female  38     1
## 3       1     3 female  26     0
## 4       1     1 female  35     1
## 5       0     3 male    35     0
## 6       0     1 male    54     0
## 7       0     3 male     2     3
## 8       1     3 female  27     0
## 9       1     2 female  14     1
## 10      1     3 female   4     1
## # ... with 704 more rows
```

We are interested in predicting the survival of the passenger (`Survived` - 1="yes", 0="no") using the variables `Pclass` (Passenger Class), `Sex`, `Age` and `SibSp` (number of sibling/spouses aboard) as predictor variables. We split the data into a training set `titanic_train` and a test set `titanic_test`. Since we have binary response variable, a logistic regression model will be fitted to the training data.

```
model_full <- glm(Survived ~ ., data = titanic_train, family = binomial)
```

A summary of the fitted model is the following

```
## # A tibble: 5 x 5
##   term          estimate std.error statistic  p.value
##   <chr>         <dbl>    <dbl>    <dbl>   <dbl>
## 1 (Intercept)    5.05     0.608     8.32 8.94e-17
## 2 Pclass        -1.09     0.155    -7.07 1.58e-12
## 3 Sexmale       -2.69     0.237   -11.3 8.96e-30
## 4 Age          -0.0422    0.00928  -4.55 5.49e- 6
## 5 SibSp         -0.409     0.135    -3.02 2.50e- 3
```

a) Write out the form of the fitted model.

The fitted model has the form

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = 5.05 - 1.09 \cdot \text{Pclass} - 2.69 \cdot \text{Sexmale} - 0.0422 \cdot \text{Age} - 0.409 \cdot \text{SibSp}$$

b) Give an **interpretation** of the estimated slope of `Sex` in terms of the odds ratio. \ Remark: Computing the odds ratio is necessary, but not sufficient. You should describe in words the meaning of the computed value.

The odds ratio for a change from female to male is equal to $e^{-2.69} \approx 0.07$, i.e. that the odds of surviving the Titanic disaster is decreased by almost 93%, given all else being equal.

c) Given the full model, we are now able to compute predictions for the test dataset.

```
titanic_test$pred_prob <- predict(model_full, newdata = titanic_test, type = "response")
```

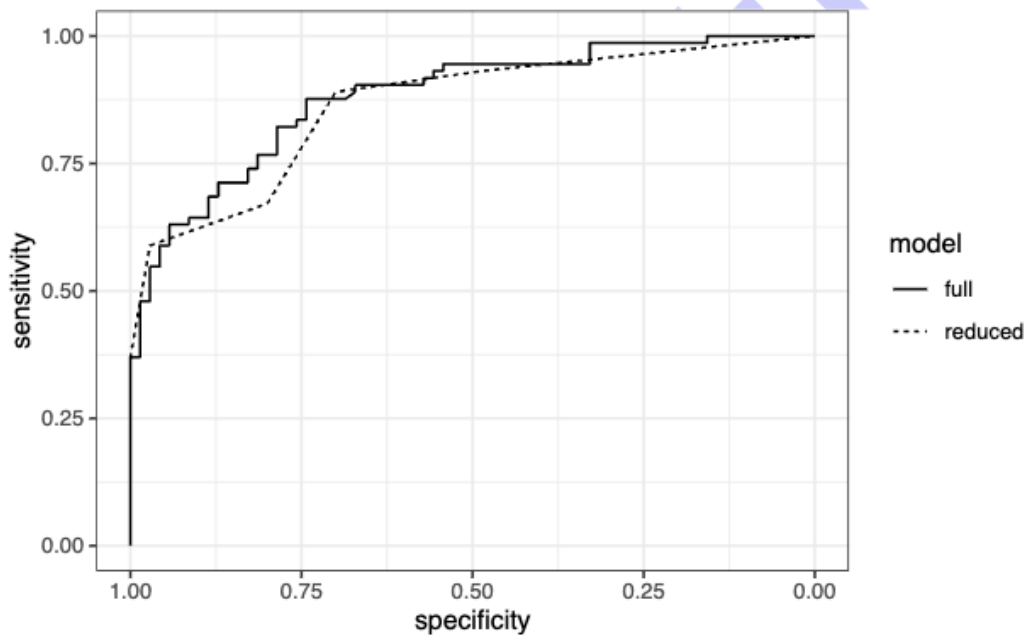
To predict survival we need a threshold for the computed probabilities of survival. Choosing a threshold of 0.5 leads to the following table

```
table(titanic_test$pred_prob > 0.5, titanic_test$Survived)
##
##      0  1
## FALSE 56 17
##  TRUE  14 56
```

Compute the specificity of this prediction.

The specificity of this prediction is equal to $\frac{56}{56+14} = 0.8$.																			
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d) Finally we compare the predictive accuracy of the model to a reduced model containing just the variables Sex and Pclass. The following graphic contains the ROC curves of the full and reduced model.



The AUC value of the full model is equal to 0.881. Decide if the reduced model has a smaller or larger AUC value. Justify your answer.

The AUC value is the area under the ROC curve. Since the ROC curve of the full model is almost always above the ROC curve of the reduced model, the AUC value of the reduced model will be smaller than 0.881.																			
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Cheat sheet

Summarizing data:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad s = \sqrt{s^2} \quad IQR = Q_3 - Q_1 \quad Q_1 - 1.5 \cdot IQR, \quad Q_3 + 1.5 \cdot IQR$$

Probability:

$$P(A^c) = 1 - P(A); \quad P(A \cap B) = P(A) \cdot P(B), \quad A, B \text{ independent}; \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)};$$

$$E(X) = \sum_{i=1}^k x_i P(X = x_i), \quad E(X) = \int_{\mathbb{R}} x \cdot f(x) dx; \quad \text{Var}(X) = E(X^2) - E(X)^2;$$

$$SD(X) = \sqrt{\text{Var}(X)}; \quad \text{Cov}(X, Y) = E(X \cdot Y) - E(X) \cdot E(Y); \quad Z = \frac{X - E(X)}{\sqrt{\text{Var}(X)}}$$

$$\text{Geom. dist.: } P(X = x) = (1 - p)^{x-1} p, \quad x \in \mathbb{N}; \quad E(X) = \frac{1}{p}, \quad SD(X) = \sqrt{\frac{1-p}{p^2}}$$

$$\text{Binom. dist.: } P(X = x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}, \quad x \in \{0, 1, 2, \dots, n\}; \quad E(X) = np, \quad SD(X) = \sqrt{np(1-p)}$$

$$\text{Neg. Binom. dist.: } P(X = x) = \frac{(x-1)!}{(k-1)!(x-k)!} p^k (1-p)^{x-k}, \quad x \in \mathbb{N};$$

$$\text{Pois. dist.: } P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x \in \mathbb{N}_0; \quad E(X) = \lambda, \quad SD(X) = \sqrt{\lambda}$$

Inference for categorical data:

$$\hat{p} \pm z^* \cdot SE_{\hat{p}} \quad SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad Z = \frac{\hat{p} - p_0}{SE_{\hat{p}}} \quad SE_0 = \sqrt{\frac{p_0(1-p_0)}{n}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \cdot SE \quad SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \quad Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{SE_0} \quad SE_0 = \sqrt{\hat{p}_{pooled}(1-\hat{p}_{pooled}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$\hat{p}_{pooled} = \frac{\sum x_{1,i} + \sum x_{2,i}}{n_1 + n_2}; \quad \chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}, \quad \text{expected} = \frac{\text{row} \times \text{column}}{\text{total}} (\text{ind.}), \quad df = k - 1 \text{ or } (r-1) \cdot (c-1)$$

Inference for numerical data:

$$\bar{x} \pm t_{df}^* \cdot SE_{\bar{x}} \quad T_{df} = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}} \quad SE_{\bar{x}} = \frac{s}{\sqrt{n}} \quad df = n - 1$$

$$(\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \cdot SE \quad T_{df} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE} \quad SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \quad df = \min(n_1 - 1, n_2 - 1)$$

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \quad df = n_1 + n_2 - 2$$

ANOVA and linear regression:

$$F = \frac{MSG}{MSE} \quad df_1 = k - 1, \quad df_2 = n - k, \text{ for } k \text{ groups}; \quad T_{df} = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\frac{MSE}{n_i} + \frac{MSE}{n_j}}} \quad df = n - k \quad \alpha^* = \alpha / K$$

$$\hat{y} = b_0 + b_1 x \quad b_1 = R_{xy} \cdot \frac{s_y}{s_x} \quad \bar{y} = b_0 + b_1 \bar{x} \quad \text{residual} = y - \hat{y}$$

$$\hat{\sigma} = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2} = \sqrt{\frac{1}{n-2} \sum (y_i - b_0 - b_1 x_i)^2}$$

$$R_{xy} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{s_x \cdot s_y} \quad T_{df} = \frac{b_1 - \beta_{1,0}}{SE_{b_1}} \quad b_1 \pm t_{df}^* \cdot SE_{b_1} \quad df = n - 2$$

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}; \quad R_{adj}^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2 / (n - k - 1)}{\sum_{i=1}^n (y_i - \bar{y})^2 / (n - 1)};$$

$$AIC = \text{constant} + n \cdot \ln \left(\sum_{i=1}^n e_i^2 \right) + 2k$$

Logistic regression:

$$\text{logit}(p_i) = \log \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i} \quad OR_j = \frac{\text{odds when } x_{i,j} = x+1}{\text{odds when } x_{i,j} = x} = e^{\beta_j}$$

$$\text{sensitivity} = P(\text{Test} + | \text{Condition} +), \quad \text{specificity} = P(\text{Test} - | \text{Condition} -)$$